Adsorption and desorption performance and stability of amine functionalized resins in Direct Air Capture

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Introduction

Greenhouse gas emissions, particularly CO_2 and CO, are major contributors that raise the surface temperature of our Earth, leading to global warming & extreme weather events. Direct Air Capture (DAC) is a promising technology to remove these anthropogenic emissions from the atmosphere by adsorption. Effective CO_2 adsorbents include materials such as Resins, MOF, activated carbon and zeolites.

Avantium R&D Solutions has created a custom high-throughput testing platform for testing materials for CO_2 adsorption. Within our adsorbent screening services, we do research and validation for a wide range of different customers. In this abstract, the results of DAC adsorption test of four different commercially used adsorption resins are tested in parallel and analyzed.

Experimental

Four different functionalized resins (100 mg each) were dried at 90 °C under nitrogen for 2 hours before starting the adsorption cycles. As pre-treatment, the materials were exposed to humid nitrogen at 23 °C. During the adsorption cycles (6x), 100 ml/min of N_2 containing 450 ppm of CO_2 at 23 °C was passed over the resins at 53% relative humidity (1500 ppm H_2O). CO_2 desorption was conducted by heating to 90 °C at a rate of 60 °C/h. CO_2 levels were monitored using an online mass spectrometer (MS).

Results

In this study, we subjected four resins to multiple cycles of adsorption and desorption. The findings highlight distinct variations in the breakthrough curve profiles, initial capacities, and deactivation rates among the resins. The upper part of Figure 1 illustrates the CO_2 breakthrough curves during the second cycle, indicating differing adsorption behaviors of the four resins due to variations in their chemical and physical properties, such as surface chemistry or pore size.

Figure 1 further shows the initial high adsorption capacity of material D compared to the other three resins, ranking as follows: D > A > B > C. The stability of the materials over five adsorption-desorption cycles was evaluated, with the stability ranking as C > A > D > B. Material C demonstrated the highest stability, maintaining its adsorption capacity over repeated cycles, while material B showed the least stability.

We will show that the differences in adsorption behavior among the materials can be attributed to variations in their chemical and physical properties, such as surface chemistry or particle size.

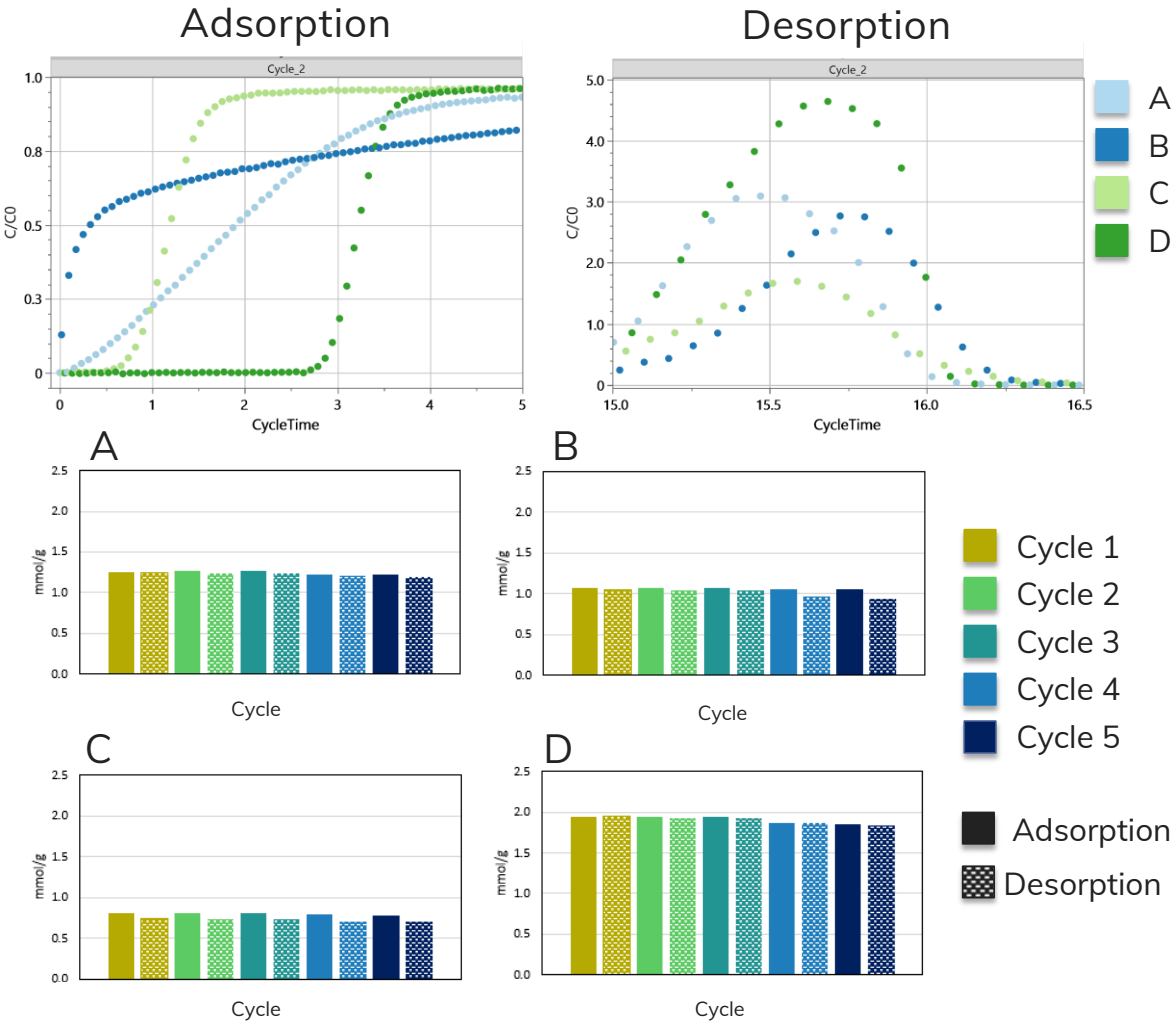

Figure 1. ***CO_2 Adsorption & Desorption Curve (upper part) and capacity (lower part) of four resins per cycle***